1. The counter which require maximum number of FF for a given mod counter is

   A. Ripple counter

   B. BCD counter

   C. Ring counter

   D. Programmed counter

   Answer: Option **C**
   Explanation:
   No answer description available for this question. Let us discuss.
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2. Four inputs A, B, C, D are fed to a NOR gate. The output of NOR gate is fed to an inverter. The output of inverter is

   A. A + B + C + D

   B. A + B + C + D

   C. ABCD

   D. ABCD

   Answer: Option **A**
   Explanation:

   Output = $\overline{\overline{A + B + C + D}}$ = A + B + C + D.

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3. The content of which of the following determines the state of the CPU at the end of the execute cycle (when the interrupt is recognized)?

   Program counter

   Processor register

   Certain status conditions

   Select the correct answer using the codes given below :

   A. 1 and 2

   B. 2 and 3

   C. 1 and 3

   D. 1, 2 and 3

   Answer: Option **C**
   Explanation:

   It is the state at which interrupt is recognized program counter and certain status condition.

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4. In a gate output is Low if and only if all inputs are High. The gate is

   A. OR

   B. AND

   C. NOR

   D. NAND

   Answer: Option **D**
   Explanation:
   No answer description available for this question. Let us discuss.
   View Answer Discuss in Forum Workspace Report

5. With on-chip decoding, *n* address lines can access

   A. 2*n* memory locations

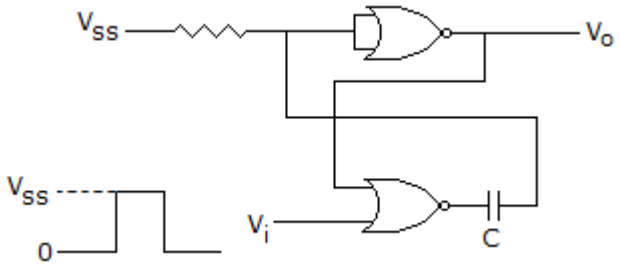
   B. $2n^2$ memory locations

   C. 2^n memory locations

   D. n^2 memory location

   Answer: Option **C**
   Explanation:
   No answer description available for this question. Let us discuss.

6. The circuit in the figure is has two CMOS-NOR gates. This circuit functions as a



   A. flip-flop

   B. schmitt trigger

   C. monostable multivibrator

   D. astable multivibrator

   Answer: Option **C**
   Explanation:

   It is simple monostable multivibrator.

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7. A 6 bit DAC uses binary weighted resistors. If MSB resistor is 20 k ohm, the value of LSB resistor is

   A. 20 k ohm

   B. 80 k ohm

   C. 320 k ohm

   D. 640 k ohm

   Answer: Option **D**
   Explanation:

   Resistances are R, 2R, 4R, 8R, 16R and 32R.

   LSB resistance = 32R = 32 x 20 = 640 K ohm.

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8. A 6 bit dual slope A/D converter uses a reference of -6v and a 1 MHz clock. It uses a fixed count of 40 (101000). Then, what will be input, if the output register shows 100111 at the end of conversion.

   A. 5.85 V

   B. 6 V

   C. 5 V

   D. 10 V

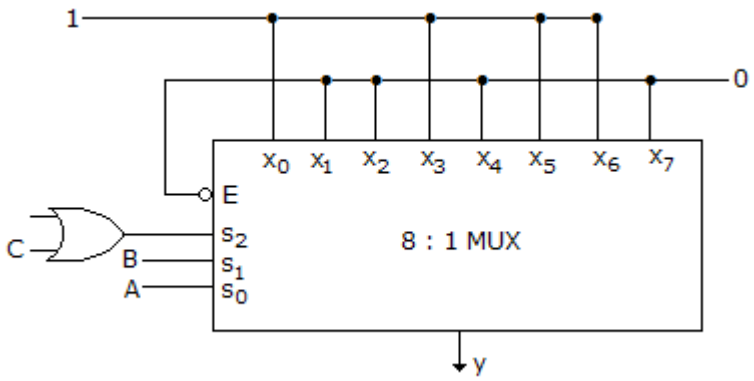
   Answer: Option **A**
   Explanation:

   $$V_i = V_R . \frac{N_2}{N_1} = 6 \times \frac{39}{40} = 5.85 \text{ V}$$

   I/P voltage and reference voltage arc of opposite polarity.

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9. In the TTL circuit in the figure, S_2 to S_0 are select lines and X_7 to X_0 are input lines. S_0 and X_0 are LSBs. The output Y is



A. Indeterminate

B. A ⊕ B

C. A ⊕ B

D. C . A ⊕ B + C . (A ⊕ B)

Answer: Option **B**

Explanation:

The MUX is made up of TTL circuit. For TTL circuit open terminal is taken high, since S_2 select line is connected to OR gate whose one terminal connected to C and the other is open (high) so OR gate output is S_2= 1 + C = 1.

S_2 = 1	S_1(B)	S0(A)	Y
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Y = $S_0 \oplus S_1$ => A⊕B.

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10. The number of accumulators in 6800 are

A. 2

B. 3

C. 1

D. 4

Answer: Option **A**

Explanation:

No answer description available for this question. Let us discuss.

11. If an analog voltage is expressed in binary using 4 bits, each successive binary count would represent

A. $\frac{1}{2}$ of total voltage

B. $\frac{1}{4}$ of total voltage

C. $\frac{1}{15}$ of total voltage

D. $\frac{1}{16}$ of total voltage

Answer: Option **C**
Explanation:

Since 1111= decimal 15, each successive binary count represent 1/15 of total voltage.

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12. In a 4 input AND gate, the total number of High outputs for 16 input states are

A. 16

B. 8

C. 4

D. 1

Answer: Option **D**
Explanation:

In only one combination are all inputs high.

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13. Symmetrical square wave of time period 100 μs can be obtained from square wave of time period 10 μs by using

A. divide by 5 circuit

B. divide by 2 circuit

C. divide by 5 circuit followed by divide by 2 circuit

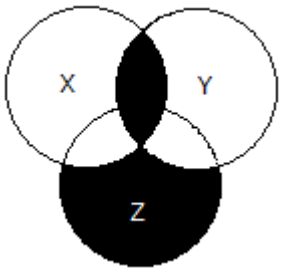
D. BCD counter

Answer: Option **C**
Explanation:

Frequency has to be divided by 10.

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14. The Boolean expression for the shaded area in the Venn diagram is



A. X + Y + Z

B. XY + Z + X YZ

C. X Y Z + XY

D. X + Y + Z

Answer: Option **C**
Explanation:
No answer description available for this question. Let us discuss.
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15. The initial state of a Mod-16 counter is 0110. After 37 clock pulses the state of counter will be

A. 1011

B. 0110

C. 010

D. 0001

Answer: Option **A**
Explanation:
After 37 clock pulses the state will be the same as after (37 - 32), *i.e.,* 5 clock pulses. 6 + 5 = 11. Hence 1011.

16. A 14 pin AND gate IC has _____ AND gates.

A. 8

B. 6

C. 4

D. 2

Answer: Option **C**
Explanation:

Each AND gates requires 3 pins, 2 inputs and one output, 1 supply pin and 1 ground pin. Hence 4 AND gates.

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17. In 8085 microprocessor, how many lines are there in data bus?

A. 6

B. 8

C. 2

D. 16

Answer: Option **B**
Explanation:
No answer description available for this question. Let us discuss.
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18. The basic shift register operations are

A. serial in - serial out

B. serial in - parallel out

C. parallel in - serial out

D. all of the above

Answer: Option **D**
Explanation:
No answer description available for this question. Let us discuss.
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19. The hexadecimal number 'A0' has the decimal value

A. 80

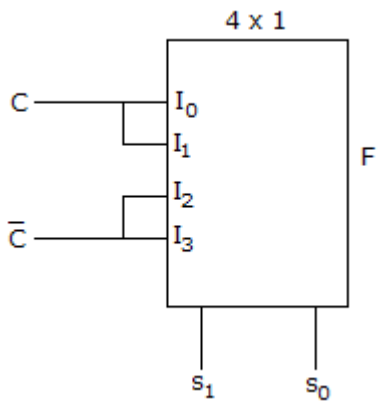
B. 256

C. 100

D. 160

Answer: Option **D**
Explanation:
'Ao' = 10 x 16^1 + 0 = 160.
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20. The logic realized by the circuit shown in figure below is



A. F = A - C

B. F = A + C

C. F = B ⊙ C

D. F = B ⊕ C

Answer: Option **D**
Explanation:
F = A B C + A BC + AB C + ABC = (AC + AC) (B + B) = A ⊕ C.

21. A 4 bit ripple counter is in 0000 state. The clock pulses are applied and then removed. The counter reads 0011. The number of clock pulses which have occurred are

A. 3

B. 3 or 19

C. 3 or 19 or 35

D. none of the above

Answer: Option **C**
Explanation:

After 16 pulses the counter resets. 0011 is 3. Hence clock pulses can be 3 or 3 + 16 or 3 + 16 + 16.

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22. TTL logic is preferred to DRL logic because

A. greater fan-out is possible

B. greater logic levels are possible

C. greater fan-in is possible

D. less power consumption is achieve

Answer: Option **A**
Explanation:
No answer description available for this question. Let us discuss.

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23. **Assertion (A):** Schottky transistors are preferred over normal transistors in digital circuits
Reason (R): A Schottky transistor when used as a switch, between cutoff and active region.

A. Both A and R are correct and R is correct explanation of A

B. Both A and R are correct but R is not correct explanation of A

C. A is true, R is false

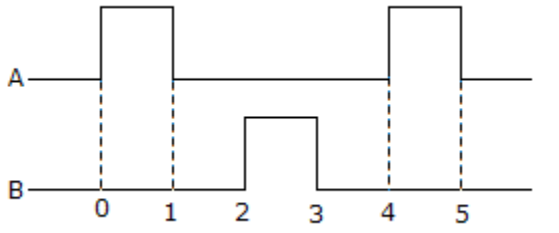
D. A is false, R is true

Answer: Option **A**
Explanation:

Schottky transistors are fast switching because saturation is avoided.

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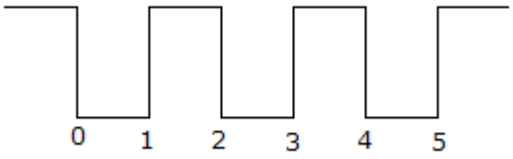
24. The inputs A and B of the given figure are applied to a two input NOR gate. The output waveform is



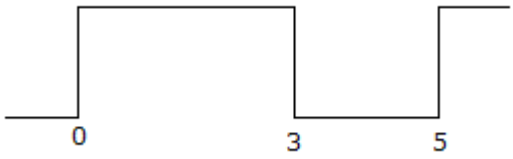
A.



B.



C.

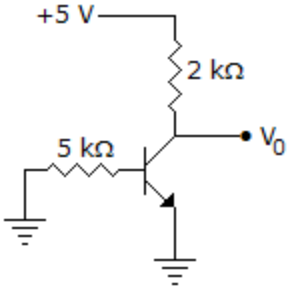


D.



Answer: Option **B**
Explanation:
No answer description available for this question. Let us discuss.
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25. In the circuit of the given figure, V_0 =



A. 5 V

B. 3.1 V

C. 2.5 V

D. 0

Answer: Option **A**
Explanation:

Transistor is off and circuit current is zero.

26. A parallel in-parallel out shift register can be used to introduce delay in digital circuits.

A. True

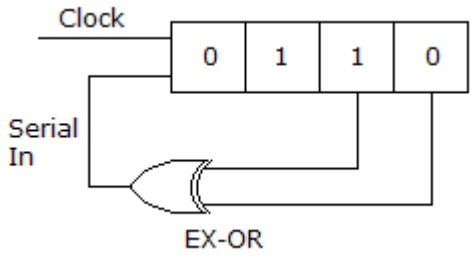
B. False

Answer: Option **B**
Explanation:

No time delay in this shift register since all bits are loaded and read simultaneously.

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27. In following figure, the initial contents of the 4-bit serial in parallel out, right shift, shift register as shown in figure are 0110. After 3 clock pulses the contents of the shift register will be



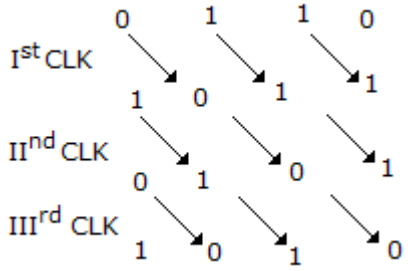
A. 0000

B. 0101

C. 1010

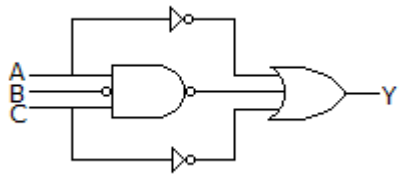
D. 1110

Answer: Option **C**
Explanation:



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28. For the logic circuit of the given figure, the minimized expression is



A. $Y = \overline{A\overline{B}C}$

B. Y = A + B + C

C. Y = A + B

D. Y = ABC

Answer: Option **A**
Explanation:

$Y = \overline{A\,\overline{B}\,C}$ + A + C = A + B + C + A + C = A + C + B

= $Y = \overline{A\,\overline{B}\,C}$

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29. Binary multiplication can be done by repeated addition.

    A. True

    B. False

    Answer: Option **A**
    Explanation:
    No answer description available for this question. Let us discuss.
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30. **Assertion (A):** In totem pole output the output impedance is low.
    Reason (R): TTL gate with active pull up should not be used in wired AND connection.

    A. Both A and R are correct and R is correct explanation of A

    B. Both A and R are correct but R is not correct explanation of A

    C. A is true, R is false

    D. A is false, R is true

    Answer: Option **B**
    Explanation:
    No answer description available for this question. Let us discuss.

31. A 4 bit synchronous counter uses flip flops with a delay time of 15 ns each. The time required for change of state is

    A. 15 ns

    B. 30 ns

    C. 45 ns

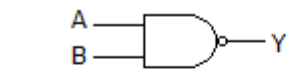
    D. 60 ns

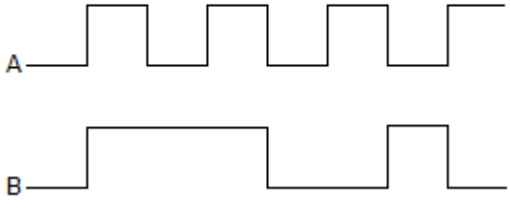
    Answer: Option **A**
    Explanation:

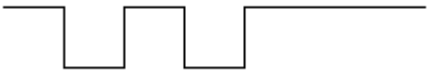
    In a synchronous counter clock input is applied to all flip flops simultaneously. Hence total delay time is 15 ns.

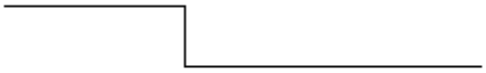
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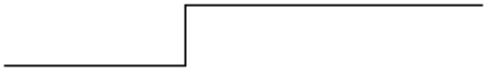
32. The inputs to a NAND gate are as shown in the given figure. The waveform of output is





A. 

B. 

C. 

D. 

Answer: Option **A**
Explanation:

NAND gate gives high output when both inputs are Low.

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33. What do the contents of instruction register specify?

A. Operand for the instruction being executed

B. Op code for the instruction being executed

C. Op code for the instruction to be executed next

D. Operand for the instruction to be executed next

Answer: Option **B**
Explanation:
No answer description available for this question. Let us discuss.
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34. In the decimal number 27, the digital 2 represents

A. 2

B. 20

C. 0.2

D. 200

Answer: Option **B**
Explanation:

2 x 10 = 20.

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35. Which of the following is true?

A. SOP is a two level logic

B. POS is a two level logic

C. Both SOP and POS are two level logic

D. Hybrid function is two level logic

Answer: Option **C**
Explanation:
No answer description available for this question. Let us discuss.

36. $E7F6_{16}$ = _____ .

A. 600000_{10}

B. 59382_{10}

C. 9382_{10}

D. 382_{10}

Answer: Option **B**
Explanation:
E7 F6 in hexadecimal = $14 \times 16^3 + 7 \times 16^2 + 15 \times 16 + 6 = 59382$ in decimal.
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37. What is the normal range of analog input voltage?

A. 0 to 1 V

B. 0 to 5 V

C. 5 to 15 V

D. 15V to 30V

Answer: Option **B**
Explanation:
No answer description available for this question. Let us discuss.

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38. **Assertion (A):** The propagation delay in ECL is minimum
Reason (R): Transistors used in ECL switch between active and cutoff regions.

A. Both A and R are correct and R is correct explanation of A

B. Both A and R are correct but R is not correct explanation of A

C. A is true, R is false

D. A is false, R is true

Answer: Option **A**
Explanation:

Since saturation is avoided switching speed is high.

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39. For the design of a sequential circuit having 9 states, minimum number of memory elements required is

A. 3

B. 4

C. 5

D. 9

Answer: Option **B**
Explanation:
$2^3 = 8$ and $2^4 = 16$. Hence minimum of 4 memory elements.
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40. ICs are

A. analog

B. digital

C. both analog and digital

D. mostly analog

Answer: Option **C**
Explanation:

Op-amp is an analog IC.

41. In a 5 x 7 dot matrix, format, to store 64 alphanumeric characters we require

A. 1120 bits

B. 2240 bits

C. 33 bits

D. 64 bits

Answer: Option **B**
Explanation:
No answer description available for this question. Let us discuss.
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42. A digital system is required to amplify a binary encoded audio signal. The user should be able to control the gain of the amplifier from a minimum to a maximum in 100 increments. The minimum number of bits required to encode, in straight binary is,

A. 8

B. 6

C. 5

D. 7

Answer: Option **D**
Explanation:
Let 6 bit required = 2^6 = 64, which indicate 64 increment.
If 7 bit = 2^7 = 128, for 100 increment, 7 bit required.
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43. A 4 bit transistor register has output voltage of high-low-high-low. The binary number stored and its decimal equivalent are

A. 0101 and 5

B. 0101 and 10

C. 1010 and 5

D. 1010 and 10

Answer: Option **D**
Explanation:

1010 in binary and 8 + 2 = 10 in decimal.

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44. **Assertion (A):** A demultiplexer cannot be used as a decoder
Reason (R): A multiplexer selects one of many outputs whereas a decoder selects on output corresponding to coded input.

A. Both A and R are correct and R is correct explanation of A

B. Both A and R are correct but R is not correct explanation of A

C. A is true, R is false

D. A is false, R is true

Answer: Option **D**
Explanation:

Demux can be used as a decoder.

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45. Hexadecimal number F is called to octal number

A. 15

B. 16

C. 17

D. 18

Answer: Option **C**
Explanation:

F = 15 in decimal notation = 17 in octal notation (8 + 7 = 15).

46. A half adder can be used only for adding

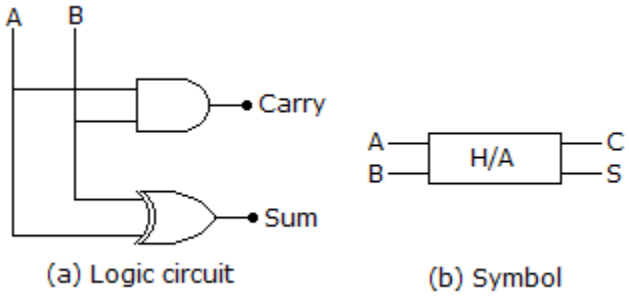
A. 1 s

B. 2 s

C. 4 s

D. 8 s

Answer: Option **A**
Explanation:



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47. In the 8421 BCD code, the decimal number 125, is written as

A. 1111101

B. 0001 0010 0101

C. 7 D

D. none of these

Answer: Option **A**
Explanation:
No answer description available for this question. Let us discuss.
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48. $110.11_2 \times 110_2 =$ _____ $_{10}$

A. 40.5

B. 30.5

C. 20.5

D. 10.5

Answer: Option **A**
Explanation:

110.11 =6.75 and 110 = 6 and 6.75 x 6 = 40.5 in decimal.

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49. Y = A + A B is the same as

A. Y = AB

B. Y = A + B

C. Y = A + B

D. Y = A + B

Answer: Option **B**
Explanation:

Verify by preparing truth table.

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50. **Assertion (A):** Different symbols are used for different gates
Reason (R): IEEE symbols are the same as traditional symbols for gates.

A. Both A and R are correct and R is correct explanation of A

B. Both A and R are correct but R is not correct explanation of A

C. A is true, R is false

D. A is false, R is true

Answer: Option **C**
Explanation:

IEEE symbols are different than traditional symbols.